

Tudor Bodea

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SUMMARY

Third-year Computer Science student at UTCN (9.8/10 average in both years). Launched Skillify, an AI learning platform that reached 24k sign-ups; placed 2nd at PoliHack with CORE-Speech, AR glasses that translate sign language; and wrote a C++ chess engine. Looking for a software engineering internship for summer 2027.

EDUCATION

2024 – 2028 **Technical University of Cluj-Napoca (UTCN)** – Computer Science, Bachelor's degree
Average 9.8/10 in Year 1 and 9.81/10 in Year 2; 10/10 in Data Structures and Algorithms, Fundamental Algorithms, Operating Systems, Computer Architecture, Databases, OOP

2020 – 2024 **Liceul de Informatică “Tiberiu Popoviciu”**, Cluj-Napoca Baccalaureate 9.7/10

PROJECTS

Skillify AI – AI learning platform

[GitHub](#)

- Built and deployed a full-stack learning platform (React, Node.js/Express, MongoDB) with an AI tutor on the OpenAI API that explains topics, quizzes users and adapts to their progress; reached 24k sign-ups.
- Implemented a freemium model with Stripe Checkout subscriptions, Google sign-in, RS256-signed JWT authentication and transactional email (Nodemailer).
- Added document, image (OCR with Tesseract.js) and YouTube summarizers that extract the text and pass it to the LLM.

CORE-Speech – 2nd place, PoliHack 2026

[GitHub](#)

- AR-glasses system that lets Deaf and hearing people talk in real time: sign language to speech, and speech to captions on the glasses.
- Handled system integration: connected sign recognition (MediaPipe, PyTorch LSTM), a local LLM (Ollama), voice cloning (OpenVoice, MeloTTS) and speech-to-text (Vosk) into one real-time pipeline.
- FastAPI edge server streaming audio over WebSockets to the ESP32-S2 firmware on the glasses (I2S microphone, display).

Chess Engine – C++, UCI protocol

[GitHub](#)

- Bitboard move generation (magic bitboards, Zobrist hashing), verified with the standard perf tests.
- Negamax alpha-beta search with iterative deepening, quiescence search, null-move pruning, MVV-LVA move ordering and a transposition table.
- Tapered PeSTO evaluation and time management; scores around 2000 Elo CCRL.

AWARDS AND COMPETITIONS

- **2nd place**, PoliHack 2026 – CORE-Speech
- **1st place**, Quantum Speed – World Quantum Day (April 2023)
- Also competed in: Cloudflight Coding Contest, AtCoder

SKILLS

Programming	C, C++, Java, Python, JavaScript, C#
Web and backend	React, Node.js, Express, FastAPI, REST, WebSockets, Stripe, OAuth, JWT
AI and ML	PyTorch, OpenCV, MediaPipe, OpenAI API, Ollama
Databases	MongoDB, Oracle Database (SQL)
Systems and hardware	Linux, Git, ESP32, VHDL, FPGA
Languages	Romanian (native), English (B2), German (A1)
Interests	Chess, climbing, cycling, hiking, reading, philosophy